

Mastering Excel: Complete Formula Practice Assignment

Data Source Vector: excel_student_practice_250.csv (250 E-Commerce Records)

Workbook Structure Guide: This comprehensive assignment guides you from basic arithmetic up to advanced lookup, conditional array summaries, and text extraction formulas. Open excel_student_practice_250.csv, immediately save it as .xlsx, and use it to solve each task below.

Part 1: Basic Math & Core Mathematical Functions

Q1 Row-by-Row Total Revenue Calculation

In column I, head the column as **Total_Revenue**. Write a foundational expression to multiply the first item row's Quantity by its corresponding Unit Price. Apply it down the full sheet.

```
=F2 * G2
```

Q2 Total Comprehensive Retail Volume (SUM)

Find a blank reference area down below your table dataset. Compute the net gross revenue generated across all 250 rows in the system.

```
=SUM(I2:I251)
```

Q3 Average Item Volume per Basket (AVERAGE)

Calculate the mean volume of items purchased per unique order placed.

```
=AVERAGE(F2:F251)
```

Q4 Tracking Order Extremes (MIN & MAX)

Determine the absolute highest single unit price logged across products, as well as the lowest unit price sold.

```
Highest: =MAX(G2:G251) | Lowest: =MIN(G2:G251)
```

Q5 Absolute Row Counter Check (COUNTA)

Verify exactly how many orders are populated inside the sheet by checking filled alphanumeric characters in the Order column.

```
=COUNTA(A2:A251)
```

Part 2: Intermediate Summary Functions & Basic Conditionals

Q6 Targeted Regional Order Counting (COUNTIF)

Isolate geographical metrics: count how many individual sales transactions came explicitly from the **West** region.

```
=COUNTIF(C2:C251, "West")
```

Q7 Category Revenue Tracking (SUMIF)

Extract the total gross revenue yielded specifically by the sale of items belonging to the **Electronics** category.

```
=SUMIF(D2:D251, "Electronics", I2:I251)
```

Q8 Isolating Segment Mean Pricing (AVERAGEIF)

Determine the average unit price spent specifically on items categorized strictly as **Clothing**.

```
=AVERAGEIF(D2:D251, "Clothing", G2:G251)
```

Q9 Single Condition Order Flagging (Logical IF)

In column J, header it **Order_Size_Class**. Write a logical evaluation rule: If a transaction's item Quantity is 4 or greater, print "Bulk". Otherwise, print "Standard".

```
=IF(F2>=4, "Bulk", "Standard")
```

Part 3: Advanced Conditional Matrix Operations (Multi-Criteria)

Q10 Multi-Criteria Summary Volumes (SUMIFS)

Find the exact accumulated total revenue generated exclusively by transactions fulfilling a double condition: occurring inside the **North** region AND involving the **Home & Kitchen** category.

```
=SUMIFS(I2:I251, C2:C251, "North", D2:D251, "Home & Kitchen")
```

Note: In SUMIFS, the column you add up always comes first!

Q11 Multi-Criteria Counter Evaluations (COUNTIFS)

Calculate how many unique orders inside the dataset were paid via **PayPal** AND contained a transaction Quantity matching exactly 1.

```
=COUNTIFS(H2:H251, "PayPal", F2:F251, 1)
```

Q12 Multi-Criteria Mean Target Extractions (AVERAGEIFS)

Evaluate financial averages for a specific combination: What is the mean total revenue of orders that originated from the **East** region using **Credit Card**?

```
=AVERAGEIFS(I2:I251, C2:C251, "East", H2:H251, "Credit Card")
```

Part 4: Data Cleanse, Mutation, & Structural Text Actions

Q13 Text Extraction from String Sequences (LEFT & RIGHT)

Let's isolate raw tracking numbers. In a new column, parse out just the numerical sequence digits from the alphanumeric Order_ID value (e.g., pulling "1001" out from "ORD-1001").

```
=RIGHT(A2, 4)
```

Q14 Combining String Fields Natively (CONCATENATE / Ampersand)

Create a descriptive localized tracking label line matching this format layout: "Product Name - Region" (e.g., "Smart Watch - East").

```
=E2 & " - " & C2
```

Q15 Standardizing Structural Casing (UPPER / LOWER)

Force all alphabetical payment method inputs in column H to display uniformly in clean, standardized capital letters.

```
=UPPER(H2)
```

Part 5: Enterprise-Grade Data Lookup & Index Frameworks

Q16 Classic Vertical Attribute Retrieval (VLOOKUP)

Assume you are building an invoice lookup card on an independent summary worksheet. If a user inputs "ORD-1005" into a target cell K2, write a lookup script to extract that order's matching **Product Name** from your main database table matrix (Columns A through I).

```
=VLOOKUP(K2, A2:I251, 5, FALSE)
```

The number 5 represents the exact position index of the Product column.

Q17 Advanced Modern Vector Matching (XLOOKUP)

Leverage modern Excel functionality to complete the same dynamic tracking fetch task: Look up an Order_ID cell reference value K2 inside your index field, and extract its exact corresponding financial **Total_Revenue** value instantly.

```
=XLOOKUP(K2, A2:A251, I2:I251)
```

Q18 Two-Dimensional Coordinate Structural Mapping (INDEX & MATCH)

Build a safe, structural matrix search mechanism that doesn't break if columns are re-ordered. Retrieve the specific **Category** description for an order code matching string value reference listed in cell K2.

```
=INDEX(D2:D251, MATCH(K2, A2:A251, 0))
```

Part 6: Financial Error Isolation & Logical Cascades

Q19 Safe Structural Formula Execution Wrappers (IFERROR)

Wrap your VLOOKUP query from Question 16 so that if someone types an invalid order number that does not exist in the dataset, Excel outputs a user-friendly alert string message reading "Order Found Blank" instead of showing an ugly #N/A cell error flag.

```
=IFERROR(VLOOKUP(K2, A2:I251, 5, FALSE), "Order Found Blank")
```

Q20 Dynamic Multi-Tier Tiering Models (Nested IFS)

Create a dynamic tier classification field in Column K headed **Revenue_Tier**. Assign tiers based on the following scale: If Total_Revenue is greater than \$200, output "Platinum Tier"; if it's greater than \$75, output "Gold Tier"; otherwise, output "Standard Tier".

```
=IFS(I2>200, "Platinum Tier", I2>75, "Gold Tier", TRUE, "Standard Tier")
```